Type errors

A type error isn't a typing error – it means your code has muddled up one type of data with another type, such as confusing numbers with strings. It's like trying to bake a cake in your fridge – it won't work, because the fridge isn't meant for baking! If you ask Python to do something impossible, don't be surprised if it won't cooperate!


```
budget = 'Fifty' * 'Five'
```

You can multiply two numbers in Python, but you can't do multiplication with strings.

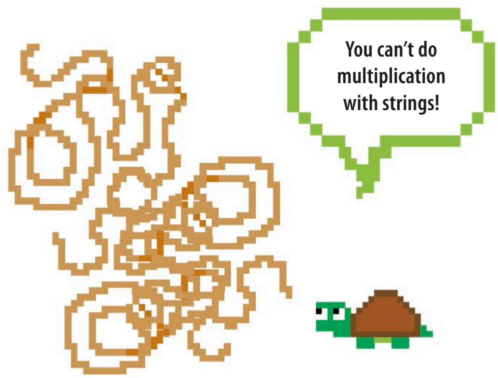
```
hot_day = '20 degrees' > 15
```

Python can't check to see if a string is greater than a number, as they are different data types.

```
list = ['a','b','c']
find_biggest_number(list)
```

This function is expecting you to give it a list of numbers, but you've given it a list of letters instead!

◁ **Examples of type errors**
Type errors occur when you ask Python to do something that doesn't make sense to it, such as multiplying with strings, comparing two completely different types of data, or telling it to find a number in a list of letters.



Name errors

A name error message appears if your code uses the name of a variable or function that hasn't yet been created. To avoid this, always define your variables and functions before you write code to use them. It's good practice to define all your functions at the top of your program.

▷ **Name errors**
A name error in this code stops Python from displaying the message "I live in Moscow". You need to create the variable `hometown` first, before you use the `print()` function.

The `print()` instruction needs to come after the variable.

```
print('I live in ' + hometown)
hometown = 'Moscow'
```